

Arjun Mauji

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Education

B.A.Sc. Mechanical Engineering, Queen's University

Kingston, Canada | 2027

- Dean's Scholar 24/25 & 25/26, Biggs – Ronald & Deanna Scholarship
- Relevant Coursework/Concepts: Fluid Dynamics, Thermodynamics, Material Sciences, Embedded Programming, Microcontrollers, Computer Architecture, Applied Control Theory

Work Experience

Teaching Assistant (Mechatronics Course), Queen's University

Sep – Dec 2025

- Co-supervised 10 mechatronics labs with over 100 students on measurement sensors and signal noise filtering
- Answered electronic measurement theory questions and troubleshoot lab circuits

Projects

Wing Geometry Analysis, Personal Project | [Report](#)

Aug 2026

- Analyzed how taper, aspect ratio, and twist affect lift and drag across 64 wing designs using **XFLR5** and **Python**
- Compared rigid and flexible wings across 6 aspect ratios to see how aeroelastic deformation affects performance

Two Stage Shifting Gearbox, Team Project | [GitHub](#)

Jan – Apr 2026

- Built a 2-stage shifting gearbox and housing to maximize torque and speed for an RC car's climb and sprint modes
- Performed **AGMA fatigue analysis** and iterated gear and shaft geometry to achieve fatigue safety factors above 1.0
- Adapted an initial steel design for 3D printing and modeled the gearbox assembly in **SolidWorks**

Signal Displayer, Personal Project | [GitHub](#)

Jul 2026

- Designed a **STM32**-based signal displayer to visualize both PWM and DC signals between 0-3.3V
- Programmed a custom 8 bit SPI driver to interact with the display module
- Built an understanding of the ARM Cortex-M architecture through manual configuration of hardware peripherals

Embedded Team, Queen's Off-Road Car Design Team (Baja)

Feb – Apr 2026

- Programmed **STM32** boards in C to monitor the vehicle transmission system's thermal condition
- Assisted in migrating the vehicle's existing **CAN** bus communication library to use **FreeRTOS**

Hardware Motion Analyzer, Personal Project | [GitHub](#)

Nov – Dec 2025

- Prototyped a low-cost (<\$40) wireless motion analyzer to compare dynamic movement flexibility between users
- Programmed 2 **IMU**-based sensor modules to capture movements by calculating joint flexion and extension angles
- Implemented **RTOS** tasking with **I2C** communication to log and filter up to 10 angle data streams concurrently
- Analyzed movement flexibility and peak angle values using **SciPy**

Skills

Mechanical: SolidWorks CAD & FEA, Reading Assembly Drawings, Ansys Fluent & Workbench, XFLR5

Embedded Systems: STM32, ESP32, FreeRTOS, CAN, SPI, I2C

Software Tools: C, Python, MATLAB (Simulink), Git/GitHub, GNU Debugger (gdb)

Electrical Tools: Multimeters, Soldering, Oscilloscopes

Spoken Languages: French (Fluent)